

MLR Practice

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2024



Animal sleep example

A researcher is doing a study on animal sleep and they are interested in whether body weight, an animal's lifespan and their feeding habits affect the amount of sleep that they get. The variables in the dataset are:

- y (TotalSleep) = the average total number of hours of sleep in a day.
- x_1 (logBodyWt) = the log of the body weight in kgs.
- x_2 (Lifespan) = animal's average lifespan.
- x_3 (Feeding) = animal's feeding habits (carnivore, omnivore and herbivore).

The correlation matrix for the continuous variables is given by:

	TotalSleep	logBodyWt	LifeSpan
TotalSleep	1.0000000	-0.5387883	-0.4102024
logBodyWt	-0.5387883	1.0000000	0.6276701
LifeSpan	-0.4102024	0.6276701	1.0000000

Interpret the correlation matrix above.

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logBodyWt	-0.5387883	1.0000000	0.6276701
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Interpret the correlation matrix above.

TotalSleep and logBodyWt - There is a moderate negative correlation between TotalSleep and logBodyWt.

TotalSleep and LifeSpan - There is a moderate negative correlation between TotalSleep and LifeSpan.

LifeSpan and logBodyWt - There is a relatively strong positive correlation between LifeSpan and logBodyWt.

Use the model output below from a regression model fit to the animal sleep data to answer the following questions:

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.79930	1.11435	12.383	< 2e-16 ***
logBodyWt	-0.37847	0.21589	-1.753	0.08585 .
LifeSpan	-0.05993	0.03442	-1.741	0.08787 .
Feedingherbivore	-4.49677	1.32862	-3.385	0.00141 **
Feedingomnivore	-0.35934	1.23145	-0.292	0.77167

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.574 on 49 degrees of freedom

Multiple R-squared: 0.4663, Adjusted R-squared: 0.4228

F-statistic: 10.71 on 4 and 49 DF, p-value: 2.583e-06

1. Give the estimated regression equation from the model output.

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
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F-statistic: 10.71 on 4 and 49 DF, p-value: 2.583e-06

1. Give the estimated regression equation from the model output.

$$\hat{y}_i = 13.799 - 0.378(\log BodyWt) - 0.060(Lifespan) - 4.497(herbivore) - 0.359(omnivore)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

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2. Interpret the Feedingherbivore and Feedingomnivore estimates.

```
Call:
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)

Residuals:
    Min       1Q   Median       3Q      Max
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Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   13.79930    1.11435   12.383  < 2e-16 ***
logBodyWt     -0.37847    0.21589   -1.753  0.08585 .
LifeSpan      -0.05993    0.03442   -1.741  0.08787 .
Feedingherbivore -4.49677    1.32862   -3.385  0.00141 **
Feedingomnivore -0.35934    1.23145   -0.292  0.77167
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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Multiple R-squared:  0.4663,    Adjusted R-squared:  0.4228
F-statistic: 10.71 on 4 and 49 DF,  p-value: 2.583e-06
```

2. Interpret the Feedingherbivore and Feedingomnivore estimates.

Feedingherbivore - On average, the total hours of sleep are 4.497 hours lower for animals who are herbivores compared to carnivores, holding all else constant.

Feedingomnivore - On average, the total hours of sleep are 0.359 hours lower for animals who are omnivores compared to carnivores, holding all else constant (but this is not significant).

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

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3. Give the estimated regression equation when Feeding habit is carnivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

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3. Give the estimated regression equation when Feeding habit is carnivore.

$$\hat{y}_i = 13.799 - 0.378(\log BodyWt) - 0.060(Lifespan)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	13.79930	1.11435	12.383	< 2e-16	***
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Multiple R-squared: 0.4663, Adjusted R-squared: 0.4228

F-statistic: 10.71 on 4 and 49 DF, p-value: 2.583e-06

4. Give the estimated regression equation when Feeding habit is omnivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
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logBodyWt	-0.37847	0.21589	-1.753	0.08585 .
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Residual standard error: 3.574 on 49 degrees of freedom

Multiple R-squared: 0.4663, Adjusted R-squared: 0.4228

F-statistic: 10.71 on 4 and 49 DF, p-value: 2.583e-06

4. Give the estimated regression equation when Feeding habit is omnivore.

$$\hat{y}_i = 13.799 - 0.378(\log BodyWt) - 0.060(Lifespan) - 0.359$$

$$\hat{y}_i = 13.440 - 0.378(\log BodyWt) - 0.060(Lifespan)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.79930	1.11435	12.383	< 2e-16 ***
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Residual standard error: 3.574 on 49 degrees of freedom

Multiple R-squared: 0.4663, Adjusted R-squared: 0.4228

F-statistic: 10.71 on 4 and 49 DF, p-value: 2.583e-06

5. Give the estimated regression equation when Feeding habit is herbivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt + LifeSpan + Feeding, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.549	-2.301	-0.545	1.686	10.767

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
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Residual standard error: 3.574 on 49 degrees of freedom

Multiple R-squared: 0.4663, Adjusted R-squared: 0.4228

F-statistic: 10.71 on 4 and 49 DF, p-value: 2.583e-06

5. Give the estimated regression equation when Feeding habit is herbivore.

$$\hat{y}_i = 13.799 - 0.378(\log BodyWt) - 0.060(Lifespan) - 4.497$$

$$\hat{y}_i = 9.302 - 0.378(\log BodyWt) - 0.060(Lifespan)$$

We fit a new model to the data and now include an interaction between logBodyWt and Feeding, the model results are shown below:

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.85473	1.16803	11.862	9.8e-16 ***
logBodyWt	-0.34102	0.31075	-1.097	0.27806
Feedingherbivore	-4.30472	1.43493	-3.000	0.00431 **
Feedingomnivore	-0.51173	1.30275	-0.393	0.69624
LifeSpan	-0.06307	0.03796	-1.662	0.10323
logBodyWt:Feedingherbivore	-0.10990	0.39848	-0.276	0.78392
logBodyWt:Feedingomnivore	0.12661	0.50634	0.250	0.80364

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.637 on 47 degrees of freedom

Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

6. Give the estimated regression equation from the model output.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
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Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

6. Give the estimated regression equation from the model output.

$$\hat{y}_i = 13.855 - 0.341(\log BodyWt) - 0.063(Lifespan) - 4.304(herbivore) - 0.512(omnivore) - 0.110(\log BodyWt \times herbivore) + 0.127(\log BodyWt \times omnivore)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
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Residuals:

	Min	1Q	Median	3Q	Max
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Coefficients:

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F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

7. Give the estimated regression equation when feeding habit is carnivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
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Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

7. Give the estimated regression equation when feeding habit is carnivore.

$$\hat{y}_i = 13.855 - 0.341(\log BodyWt) - 0.063(Lifespan)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

	Min	1Q	Median	3Q	Max
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Coefficients:

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Residual standard error: 3.637 on 47 degrees of freedom

Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

8. Give the estimated regression equation when feeding habit is herbivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
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Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

8. Give the estimated regression equation when feeding habit is herbivore.

$$\hat{y}_i = 13.855 - 0.341(\log BodyWt) - 0.063(Lifespan) - 4.304 - 0.110(\log BodyWt)$$

$$\hat{y}_i = 9.551 - 0.451(\log BodyWt) - 0.063(Lifespan)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
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Coefficients:

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F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

9. Give the estimated regression equation when feeding habit is omnivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.85473	1.16803	11.862	9.8e-16 ***
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Feedingomnivore	-0.51173	1.30275	-0.393	0.69624
LifeSpan	-0.06307	0.03796	-1.662	0.10323
logBodyWt:Feedingherbivore	-0.10990	0.39848	-0.276	0.78392
logBodyWt:Feedingomnivore	0.12661	0.50634	0.250	0.80364

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Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

9. Give the estimated regression equation when feeding habit is omnivore.

$$\hat{y}_i = 13.855 - 0.341(\log BodyWt) - 0.063(Lifespan) - 0.512 + 0.127(\log BodyWt)$$

$$\hat{y}_i = 13.343 - 0.214(\log BodyWt) - 0.063(Lifespan)$$

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.85473	1.16803	11.862	9.8e-16 ***
logBodyWt	-0.34102	0.31075	-1.097	0.27806
Feedingherbivore	-4.30472	1.43493	-3.000	0.00431 **
Feedingomnivore	-0.51173	1.30275	-0.393	0.69624
LifeSpan	-0.06307	0.03796	-1.662	0.10323
logBodyWt:Feedingherbivore	-0.10990	0.39848	-0.276	0.78392
logBodyWt:Feedingomnivore	0.12661	0.50634	0.250	0.80364

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.637 on 47 degrees of freedom

Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

10. Interpret the logBodyWt:Feedingherbivore interaction estimate.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	13.85473	1.16803	11.862	9.8e-16	***
logBodyWt	-0.34102	0.31075	-1.097	0.27806	
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.637 on 47 degrees of freedom

Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

10. Interpret the logBodyWt:Feedingherbivore interaction estimate .

It is the change in the slope estimate for logBodyWt when an animal is a herbivore compared to a carnivore. So the slope for logBodyWt decreases by 0.109 when an animal is a herbivore compared to carnivore.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	13.85473	1.16803	11.862	9.8e-16	***
logBodyWt	-0.34102	0.31075	-1.097	0.27806	
Feedingherbivore	-4.30472	1.43493	-3.000	0.00431	**
Feedingomnivore	-0.51173	1.30275	-0.393	0.69624	
LifeSpan	-0.06307	0.03796	-1.662	0.10323	
logBodyWt:Feedingherbivore	-0.10990	0.39848	-0.276	0.78392	
logBodyWt:Feedingomnivore	0.12661	0.50634	0.250	0.80364	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.637 on 47 degrees of freedom

Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

11. Interpret the logBodyWt:Feedingomnivore interaction estimate.

Call:

```
lm(formula = TotalSleep ~ logBodyWt * Feeding + LifeSpan, data = animalsleep_sub)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.3976	-2.2656	-0.6301	2.0498	10.8377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.85473	1.16803	11.862	9.8e-16 ***
logBodyWt	-0.34102	0.31075	-1.097	0.27806
Feedingherbivore	-4.30472	1.43493	-3.000	0.00431 **
Feedingomnivore	-0.51173	1.30275	-0.393	0.69624
LifeSpan	-0.06307	0.03796	-1.662	0.10323
logBodyWt:Feedingherbivore	-0.10990	0.39848	-0.276	0.78392
logBodyWt:Feedingomnivore	0.12661	0.50634	0.250	0.80364

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.637 on 47 degrees of freedom

Multiple R-squared: 0.4698, Adjusted R-squared: 0.4021

F-statistic: 6.94 on 6 and 47 DF, p-value: 2.53e-05

11. Interpret the logBodyWt:Feedingherbivore interaction estimate .

It is the change in the slope estimate for logBodyWt when an animal is an omnivore compared to carnivore. So the slope for logBodyWt increases by 0.127 when an animal is an omnivore compared to carnivore.